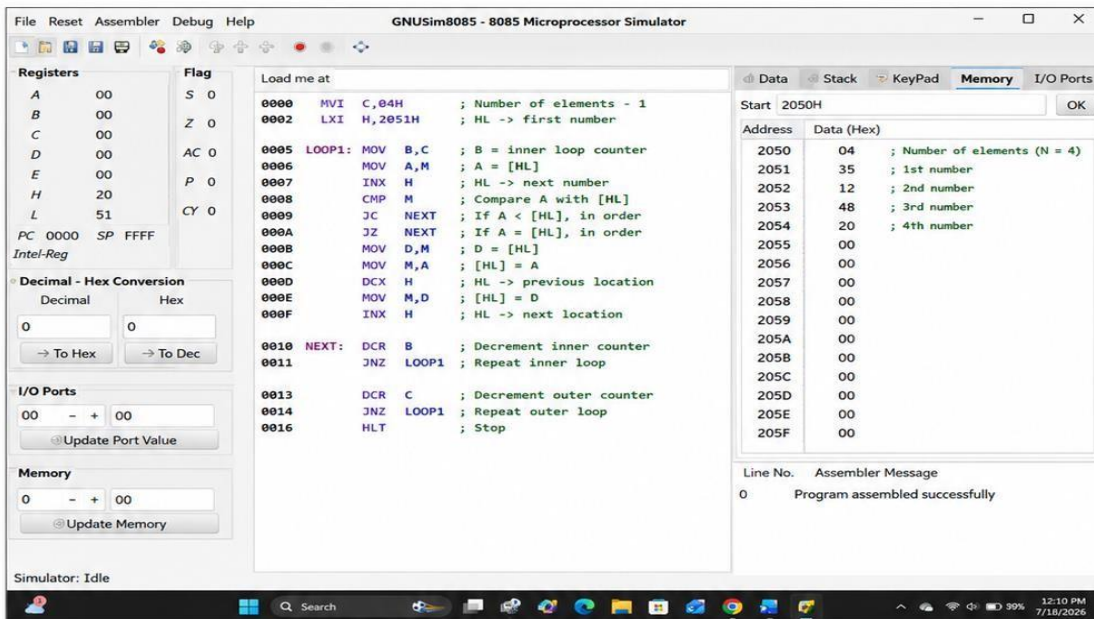


Name: Nikhil Kumar Reddy
Kadiveti

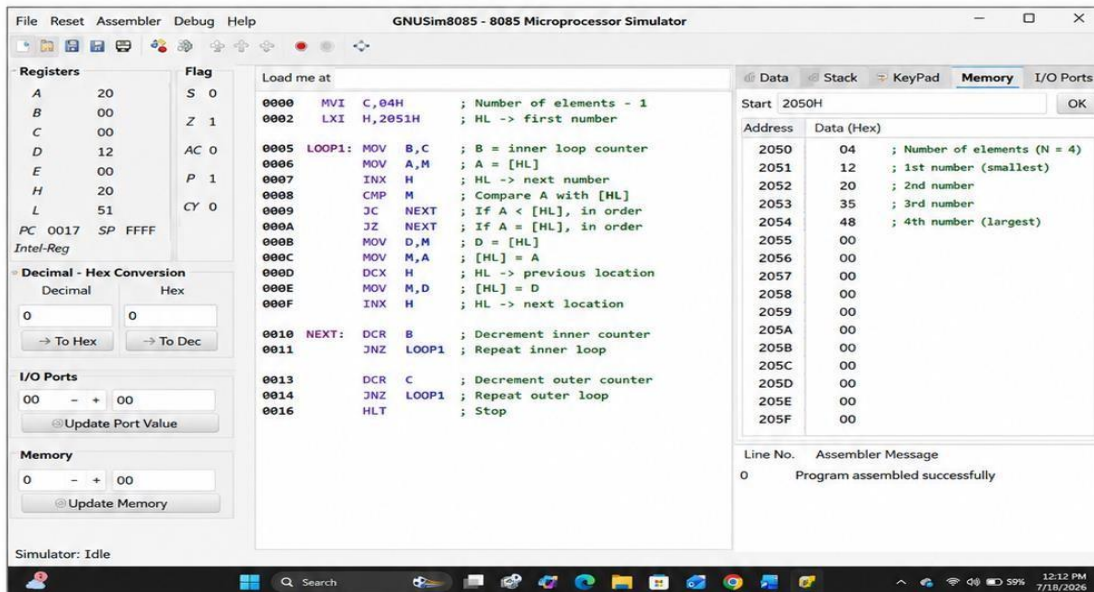
Regno :192425031

16. Write an assembly language program to arrange numbers in Ascending order.

Program 16: Arrange Numbers in Ascending Order (8085)



Before Execution (Unsorted Numbers)



After Execution (Sorted Numbers in Ascending Order)

Observation:

Before execution, the numbers were 35H, 12H, 48H, 20H.

After execution, the numbers are arranged in ascending order as 12H, 20H, 35H, 48H.

17. Write an assembly language program to arrange numbers in Descending order.

17. Arrange Numbers in Descending Order (8085)

Program : To arrange numbers in Descending order.

Before Execution (Unsorted Numbers)

Registers: A: 00, B: 00, C: 00, D: 00, E: 00, H: 20, L: 51, PC: 0000, SP: FFFF, Int-Reg: 0000, SP: FFFF, Flag: S: 0, Z: 0, AC: 0, P: 0, CY: 0.

Memory (Address 2050H):

Address	Data (Hex)	Comment
2050	04	Number of elements (N = 4)
2051	35	1st number
2052	12	2nd number
2053	48	3rd number
2054	20	4th number
2055	00	
2056	00	
2057	00	
2058	00	

Assembler Message: Program assembled successfully.

After Execution (Sorted Numbers in Descending Order)

Registers: A: 48, B: 00, C: 00, D: 20, E: 00, H: 20, L: 55, PC: 0017, SP: FFFF, Int-Reg: 0017, SP: FFFF, Flag: S: 0, Z: 0, AC: 0, P: 0, CY: 1.

Memory (Address 2050H):

Address	Data (Hex)	Comment
2050	04	Number of elements (N = 4)
2051	48	1st number (largest)
2052	35	2nd number
2053	20	3rd number
2054	12	4th number (smallest)
2055	00	
2056	00	
2057	00	
2058	00	

Assembler Message: Program assembled successfully.

Observation:

Before execution, the numbers were : 35H, 12H, 48H, 20H.

After execution, the numbers are arranged in Descending order as : 48H, 35H, 20H, 12H.

18. Write an assembly language program to find the largest number in an array.

18. Write an assembly language program to find the largest number in an array (8085).

Before Execution

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 00, B 00, C 00, D 00, E 00, H 00, L 00, PC 0000, SP FFFF, Int-Reg 00. Flags: S 0, Z 0, AC 0, P 0, CY 0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Port 0), 00 (Port 1).

Memory: 0 (Address), 00 (Data).

Load me at:

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A = first element (assume largest)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A >= [HL], no change
200B MOV A,M ; Else, A = [HL] (new largest)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store largest number at 2060H
2013 HLT ; Stop

```

Start 2050H

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	; Location to store largest number

Line No. Assembler Message
0 Program assembled successfully

Simulator: Idle

After Execution

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 8A, B 00, C 00, D 00, E 00, H 20, L 56, PC 2014, SP FFFF, Int-Reg 00. Flags: S 0, Z 0, AC 0, P 0, CY 0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Port 0), 00 (Port 1).

Memory: 0 (Address), 00 (Data).

Load me at:

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A = first element (assume largest)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A >= [HL], no change
200B MOV A,M ; Else, A = [HL] (new largest)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store largest number at 2060H
2013 HLT ; Stop

```

Start 2050H

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	8A	; Largest number stored here

Line No. Assembler Message
0 Program assembled successfully

Simulator: Idle

Input (Before Execution) :
Number of elements (N) = 5
Array elements (starting from 2051H) = 13H, 2FH, 07H, 8AH, 3EH

Output (After Execution) :
Largest number = 8AH (stored at memory location 2060H)

19. Write an assembly language program to find the Minimum number in an array.

19. Write an Assembly Language Program to Find the Minimum Number in an Array (8085).

Before Execution (Unsorted Array)

Registers: A=00, B=00, C=00, D=00, E=00, H=20, L=51, PC=2000, SP=FFFF, Int-Reg=00. Flag: S=0, Z=0, AC=0, P=0, CY=0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Data), 00 (Address).

Memory: 0 (Data), 00 (Address).

Program Code:

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A <- first element (assume minimum)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A <= [HL], no change
200B MOV A,M ; Else, A <- [HL] (new minimum)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store minimum at 2060H
2013 HLT ; Stop

```

Memory Dump:

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	; Location to store minimum

Line No. 0 Assembler Message: Program assembled successfully

After Execution (Minimum Number Stored)

Registers: A=07, B=00, C=00, D=00, E=00, H=20, L=56, PC=2014, SP=FFFF, Int-Reg=00. Flag: S=0, Z=0, AC=0, P=0, CY=0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Data), 00 (Address).

Memory: 0 (Data), 00 (Address).

Program Code:

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A <- first element (assume minimum)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A <= [HL], no change
200B MOV A,M ; Else, A <- [HL] (new minimum)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store minimum at 2060H
2013 HLT ; Stop

```

Memory Dump:

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	07	; Minimum number stored here

Line No. 0 Assembler Message: Program assembled successfully

Input (Before Execution) :
Number of elements (N) = 5
Array elements (starting from 2051H) = 13H, 2FH, 07H, 8AH, 3EH
Output (After Execution) :
Minimum number = 07H (stored at memory location 2060H)

20. Write an assembly language program to find the LCM of 2 numbers

20. Write an Assembly Language Program to Find the LCM of Two Numbers (8085).

Before Execution (Input)

After Execution (LCM Stored)

Input (Before Execution) : First Number = 02H (2), Second Number = 03H (3)

Output (After Execution) : LCM = 06H (6)

LCM is stored at memory location 2050H.

21. Write an assembly language program to find the GCD of 2 numbers.

21. Write an Assembly Language Program to Find the GCD of Two Numbers (8085).

Before Execution (Input)

Registers: A 00, B 00, C 00, D 00, E 00, H 20, L 50, PC 2000, SP FFFF, Int-Reg 00. Flags: S 0, Z 0, AC 0, P 0, CY 0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Port 0) to 00 (Port 1).

Memory: 0 (Memory) to 00 (Memory).

Program Code:

```

2000 MVI B,15H ; First number = 15H (21)
2002 MVI C,09H ; Second number = 09H (9)
2004 MOV A,B ; A = First number
2005 MOV D,C ; D = Second number
2006 LOOP: CMP D ; Compare A and D
2007 JZ DONE ; If equal, GCD found
2009 JC L1 ; If A < D, go to L1
200A SUB D ; A = A - D
200B JMP LOOP ; Repeat loop
200D L1: MOV A,D ; D = D - B
200E SUB B ; D = D - B
200F MOV D,A ; Update D
2010 JMP LOOP
2013 DONE: MOV E,A ; GCD in A -> store in E
2014 STA 2050H ; Store GCD at 2050H
2017 HLT ; Stop

```

Memory Dump:

Address	Data (Hex)	Comment
2050	00	; Location to store GCD
2051	15	; First number = 15H (21)
2052	09	; Second number = 09H (9)
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	

Line No. Assembler Message
0 Program assembled successfully

After Execution (GCD Stored)

Registers: A 03, B 15, C 09, D 03, E 03, H 20, L 50, PC 2018, SP FFFF, Int-Reg 00. Flags: S 0, Z 1, AC 0, P 1, CY 0.

Decimal - Hex Conversion: 0 (Decimal) to 0 (Hex).

I/O Ports: 00 (Port 0) to 00 (Port 1).

Memory: 0 (Memory) to 00 (Memory).

Program Code:

```

2000 MVI B,15H ; First number = 15H (21)
2002 MVI C,09H ; Second number = 09H (9)
2004 MOV A,B ; A = First number
2005 MOV D,C ; D = Second number
2006 LOOP: CMP D ; Compare A and D
2007 JZ DONE ; If equal, GCD found
2009 JC L1 ; If A < D, go to L1
200A SUB D ; A = A - D
200B JMP LOOP ; Repeat loop
200D L1: MOV A,D ; D = D - B
200E SUB B ; D = D - B
200F MOV D,A ; Update D
2010 JMP LOOP
2013 DONE: MOV E,A ; GCD in A -> store in E
2014 STA 2050H ; Store GCD at 2050H
2017 HLT ; Stop

```

Memory Dump:

Address	Data (Hex)	Comment
2050	03	; GCD = 03H (3)
2051	15	; First number = 15H (21)
2052	09	; Second number = 09H (9)
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	

Line No. Assembler Message
0 Program assembled successfully

Input (Before Execution) : First Number = 15H (21), Second Number = 09H (9)

Output (After Execution) : GCD = 03H (3)

GCD is stored at memory location 2050H.

22. Write an assembly language program to find factorial of n in the given number.

22. Write an Assembly Language Program to Find Factorial of N in the Given Number (8085).

Before Execution (Input)

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The registers are initialized as follows:

Register	Value	Flag
A	00	S 0
B	00	Z 0
C	00	AC 0
D	00	P 0
E	00	CY 0
H	20	
L	50	
PC	2000	SP FFFF
Int-Reg	00	

The memory contains the input value 05 at address 2050H. The program is assembled successfully.

After Execution (Factorial Stored)

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface after execution. The registers are updated as follows:

Register	Value	Flag
A	00	S 0
B	78	Z 1
C	00	AC 0
D	00	P 1
E	00	CY 0
H	00	
L	78	
PC	2027	SP FFFF
Int-Reg	00	

The memory contains the factorial result 120 (0078H) at address 2051H and 00 at address 2052H. The program is assembled successfully.

Input (Before Execution) : N = 05 (5)
 Output (After Execution) : Factorial of 5 = 120 = 0078H
 Result stored at memory location 2051H (Low byte) and 2052H (High byte).

23. Write a program to convert Decimal number to Hexadecimal number.

23. Write a program to convert Decimal number to Hexadecimal number (8085).

Before Execution (Input)

File Reset Assembler Debug Help

GNUSim8085 - 8085 Microprocessor Simulator

Registers: A 00, B 00, C 00, D 00, E 00, H 00, L 00, PC 2000, SP FFFF, Int-Reg 00. Flag: S 0, Z 0, AC 0, P 0, CY 0.

Decimal - Hex Conversion: Decimal 0, Hex 0. Buttons: → To Hex, → To Dec.

I/O Ports: 00 - + 00. Update Port Value.

Memory: 0 - + 00. Update Memory.

Load me at:

```

2000 MVI A,4DH ; Load decimal number (example : 77)
2002 MVI B,00H ; B will hold high nibble (MSN)
2004 MVI C,00H ; C will hold low nibble (LSN)
2006 MOV B,A ; Copy number to B
2007 RRC ; Rotate right 4 times to get high nibble
2008 RRC
2009 RRC
200A RRC
200B ANI 0FH ; Mask lower 4 bits (High nibble)
200D STA 2050H ; Store high nibble at 2050H
2010 MOV A,B ; Get original number back
2011 MOV A,4BH ; (Reload original for safety)
2013 ANI 0FH ; Mask higher 4 bits (Low nibble)
2015 STA 2051H ; Store low nibble at 2051H
2018 HLT ; Stop

```

Memory window: Address 2050 Data 00 Comment ; High nibble will be stored, Address 2051 Data 00 Comment ; Low nibble will be stored.

Line No. Assembler Message: 0 Program assembled successfully.

Simulator: Idle

After Execution (Hexadecimal Result Stored)

File Reset Assembler Debug Help

GNUSim8085 - 8085 Microprocessor Simulator

Registers: A 0D, B 04, C 0D, D 00, E 00, H 00, L 00, PC 2018, SP FFFF, Int-Reg 00. Flag: S 0, Z 0, AC 0, P 1, CY 0.

Decimal - Hex Conversion: Decimal 0, Hex 0. Buttons: → To Hex, → To Dec.

I/O Ports: 00 - + 00. Update Port Value.

Memory: 0 - + 00. Update Memory.

Load me at:

```

2000 MVI A,4DH ; Load decimal number (example : 77)
2002 MVI B,00H ; B will hold high nibble (MSN)
2004 MVI C,00H ; C will hold low nibble (LSN)
2006 MOV B,A ; Copy number to B
2007 RRC ; Rotate right 4 times to get high nibble
2008 RRC
2009 RRC
200A RRC
200B ANI 0FH ; Mask lower 4 bits (High nibble)
200D STA 2050H ; Store high nibble at 2050H
2010 MOV A,B ; Get original number back
2011 MOV A,4BH ; (Reload original for safety)
2013 ANI 0FH ; Mask higher 4 bits (Low nibble)
2015 STA 2051H ; Store low nibble at 2051H
2018 HLT ; Stop

```

Memory window: Address 2050 Data 04 Comment ; High nibble (MSN) = 4, Address 2051 Data 0D Comment ; Low nibble (LSN) = D.

Line No. Assembler Message: 0 Program assembled successfully.

Simulator: Idle

Input (Decimal number) : 77 (4DH)
Output (Hexadecimal) : 4D (High nibble = 04H, Low nibble = 0DH)
Result stored at 2050H = 04H and 2051H = 0DH.